

Dynamical Activation of a Transition Scale in Galactic Rotation Curves

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Abstract

This work examines the structure of galactic rotation curves with a focus on the transition scale q that governs the radial form of the excess acceleration component.

While previous studies have established the radial acceleration relation (MDAR/RAR) as a tight empirical scaling law, the origin of its residual structure and the diversity of rotation curve shapes remain open questions. In particular, it is unclear whether parameters used in phenomenological fits correspond to physically meaningful quantities or simply reflect continuous variations in galaxy properties.

The analysis shows that q is neither continuously distributed nor predictable from global baryonic quantities such as surface density or luminosity. Instead, the data are better described by a two-component structure, with most galaxies remaining close to $q \approx 0$ and a smaller subset exhibiting finite values.

By introducing a dynamical measure of profile structure, the turnover strength T , this separation can be described as a threshold-driven activation process. A critical value $T_c \approx 4.3$ is identified, above which a finite transition scale is systematically activated.

This behavior is interpreted as the empirical manifestation of a regime-dependent dynamical organization. The results further indicate that residuals in the MDAR/RAR are not purely stochastic, but exhibit a systematic dependence on the dynamical regime.

Taken together, these findings support the interpretation of q as a dynamically activated quantity that reflects global structural reorganization, providing a new perspective on the origin of diversity in galactic rotation curves.

Keywords: galactic rotation curves; dark matter effect; MDAR; BTFR; dynamical activation; galactic dynamics; SPARC; Universal Quantum Foam Hypothesis; UQSH

1. Introduction

Galactic rotation curves provide one of the most direct probes of the dynamical structure of galaxies. Observations show that the gravitational acceleration inferred from the motion of stars and gas systematically exceeds the contribution expected from baryonic matter alone. This discrepancy is commonly referred to as the dark matter effect [6].

A major empirical development in this context is the discovery of the radial acceleration relation (RAR), or equivalently the mass discrepancy acceleration relation (MDAR), which reveals a tight correlation between observed and baryonic acceleration across a wide range of galaxies [1, 8]. While this relation is remarkably tight, its residual structure and the diversity of individual rotation curves remain subjects of ongoing investigation.

Recent studies have emphasized that rotation curves exhibit significant structural diversity, even among systems with similar global properties [19, 21]. This raises the question of whether the observed scatter reflects purely stochastic variations, or whether it encodes additional dynamical structure that is not captured by continuous scaling relations alone.

In many phenomenological descriptions, the radial structure of rotation curves is characterized using functional forms that include a transition scale. However, such parameters are typically treated as free fitting quantities without a clear dynamical interpretation.

The central objective of this work is to examine whether the transition scale q can instead be understood as an emergent quantity linked to the internal dynamical structure of the system. Rather than assuming a specific theoretical model, the analysis is carried out in a data-driven way, focusing on the radial structure of acceleration profiles.

The key result is that q does not behave as a continuously varying parameter. Instead, the data separate into two distinct regimes: a dominant population with $q \approx 0$, and a smaller subset with finite values. This separation is found to be strongly correlated with a global measure of profile structure, the turnover strength T .

Based on this observation, the transition scale is interpreted as a dynamically activated quantity. A threshold-like behavior is identified, with a critical turnover value above which a finite scale emerges. This suggests that the diversity of rotation curves may be understood in terms of regime-dependent dynamics rather than purely continuous variation.

The structure of this paper is as follows. Section 2 describes the data and methodology. Section 3 analyzes the empirical structure of the acceleration profiles. Section 4 introduces the activation framework. Sections 5–8 examine the implications for dynamical scaling relations and large-scale phenomena. Section 9 discusses the results, and Section 10 concludes.

2. Data and Methodology

2.1. Data Sets

The primary data set used in this work is the SPARC sample [2], which provides high-quality rotation curves together with a consistent baryonic decomposition into gas, stellar disk, and bulge components. This allows a direct separation between the observed acceleration and the baryonic contribution, which is essential for constructing the excess dynamical component.

In addition, I consider independent rotation curve data from the THINGS and LITTLE THINGS surveys [5, 14]. These data sets typically provide observed rotation velocities but lack a fully consistent baryonic decomposition. They are therefore used as an independent test of the dynamical structure, rather than for a direct reconstruction of the excess term.

To further test robustness, all available data were combined into a unified corpus comprising several hundred galaxies from different surveys. Due to the heterogeneous nature of these data, only subsets

with sufficient radial resolution and physical consistency were used for quantitative analysis.

The activation relation between T and q should not be interpreted as a purely predictive model in the strict sense, since both quantities are derived from the same underlying profiles. Instead, the result should be understood as an empirical structural relation within the data, whose predictive power requires further validation on independent datasets.

2.2. Acceleration Profiles

For galaxies with full baryonic information, the observed acceleration is defined as

$$g_{\text{obs}}(r) = \frac{V_{\text{obs}}^2(r)}{r}. \quad (1)$$

The baryonic contribution is given by

$$g_{\text{bar}}(r) = \frac{V_{\text{gas}}^2(r) + V_{\text{disk}}^2(r) + V_{\text{bul}}^2(r)}{r}. \quad (2)$$

The excess component is then defined as

$$g_{\text{add}}(r) = g_{\text{obs}}(r) - g_{\text{bar}}(r). \quad (3)$$

For data sets without baryonic decomposition, an effective dynamical quantity is constructed directly from the observed rotation curve,

$$g_{\text{eff}}(r) = \frac{V_{\text{rot}}^2(r)}{r}. \quad (4)$$

This allows me to probe the large-scale structure of the dynamics without introducing additional modeling assumptions.

2.3. Dynamical Descriptors

To quantify the global structure of the acceleration profile, we introduce the logarithmic slope

$$s(r) = \frac{d \log g}{d \log r}. \quad (5)$$

The inner and outer slopes are defined as averages over the inner and outer regions of the profile:

$$s_{\text{in}} = \langle s(r) \rangle_{\text{inner}}, \quad s_{\text{out}} = \langle s(r) \rangle_{\text{outer}}. \quad (6)$$

From these, we define the turnover strength

$$T = s_{\text{in}} - s_{\text{out}}. \quad (7)$$

This quantity measures how strongly the profile changes between the inner and outer regions and serves as a global descriptor of dynamical reorganization.

To capture higher-order structure, we define curvature measures from the second logarithmic derivative:

$$\kappa(r) = \frac{d^2 \log g}{d(\log r)^2}. \quad (8)$$

From this, I define

$$C_{\text{std}} = \text{std}[\kappa(r)], \quad C_{\text{max}} = \max |\kappa(r)|. \quad (9)$$

2.4. Activation Model

The central hypothesis of this work is that the transition scale q is not an arbitrary parameter, but emerges dynamically from the structure of the acceleration profile.

I model this behavior using an activation function

$$q = q_{\text{max}} \sigma\left(\frac{a T - S_0}{w}\right), \quad (10)$$

where $\sigma(x) = (1 + e^{-x})^{-1}$ is a sigmoid function.

The parameters a , S_0 , w , and q_{max} are determined once from the SPARC sample and then kept fixed for all subsequent analyses.

2.5. Model Fit

Given a value of q , the radial dependence of the acceleration is approximated by

$$g(r) \approx A_0 \frac{r}{(r + q)^p}. \quad (11)$$

The normalization A_0 is determined analytically via least squares, while the exponent p is obtained by scanning over a finite range and selecting the value that minimizes the logarithmic error.

2.6. Error Metric

Model performance is quantified using the logarithmic mean squared error

$$\text{MSE}_{\log} = \langle (\log g_{\text{data}} - \log g_{\text{model}})^2 \rangle. \quad (12)$$

This ensures that deviations across different radial scales are treated consistently.

2.7. Validation Strategy

The activation model is calibrated on the SPARC dataset and then applied without modification to independent data sets.

This approach avoids overfitting and allows me to test whether the observed activation behavior represents a general property of galactic dynamics.

3. Empirical Structure of the Acceleration Profiles

3.1. Observed Excess Structure

After constructing the acceleration profiles, I analyze the excess component

$$g_{\text{add}}(r) = g_{\text{obs}}(r) - g_{\text{bar}}(r) \quad (13)$$

for each galaxy in the SPARC sample.

Across the full dataset, a consistent qualitative structure emerges. The excess term typically shows:

- a rise in the inner region,
- a turnover or flattening at intermediate radii,
- and a gradual decline or saturation toward larger radii.

This behavior differs significantly from simple monotonic profiles and already indicates that the dynamics cannot be described by a single global scaling relation.

3.2. Normalization and Shape Function

To compare galaxies of different sizes and amplitudes, I introduce a dimensionless normalization.

The radial coordinate is scaled as

$$x = \frac{r}{R}, \quad (14)$$

where R denotes a characteristic radius of the system, and the excess term is normalized as

$$F(x) = \frac{C(r)}{C_{\text{max}}}, \quad (15)$$

where $C(r) = V_{\text{obs}}^2(r) - V_{\text{bar}}^2(r)$.

This removes the overall amplitude and isolates the shape of the profile.

Empirically, the normalized profiles are well described by a function of the form

$$F(x) \propto x^p e^{-x/q}. \quad (16)$$

This functional form captures both the inner rise and the outer decline with only two parameters.

3.3. Deviation from Simple Scaling

A key question is whether the parameter q can be expressed as a simple function of global baryonic quantities such as surface density or luminosity.

To test this, I attempted to model q as a global scaling relation of the form

$$q = f(\Sigma, L). \quad (17)$$

However, the resulting correlation is extremely weak, with a coefficient of determination

$$R^2 \approx 0.03. \quad (18)$$

This indicates that q cannot be predicted from baryonic quantities alone.

3.4. *Implication*

At this stage, the data already suggest that the transition scale q is not a simple structural parameter tied directly to mass or luminosity.

Instead, it appears to encode additional information about the internal dynamical structure of the system.

This observation motivates a closer analysis of the radial structure of the acceleration profiles in the following sections.

4. **Emergence of Distinct Dynamical Regimes**

4.1. *Empirical Separation of Regimes*

When analyzing the distribution of the transition scale q across the sample, I find that the values do not form a continuous distribution.

Instead, the data separate into two empirically distinguishable regimes:

- a large population with $q \approx 0$,
- and a much smaller population with finite values $q > 0$.

This separation is not imposed by the model, but emerges directly from the data.

The difference between these two populations spans several orders of magnitude, indicating that the transition scale is not gradually varying, but effectively switches on only under specific conditions.

4.2. *Connection to Profile Structure*

To understand this behavior, I examine the radial structure of the acceleration profiles.

I find that galaxies with $q \approx 0$ exhibit smooth, monotonic profiles without a pronounced structural transition.

In contrast, galaxies with finite q show a clear turnover in the acceleration profile, corresponding to a change between inner and outer behavior.

This suggests that the presence of a finite transition scale is directly linked to the global structure of the profile.

4.3. Interpretation

At this stage, the data indicate that the two regimes are not independent classes, but arise from a common underlying mechanism.

The key observation is that the transition scale appears only when the profile exhibits sufficient structural reorganization.

This motivates the interpretation that the system passes between two distinct dynamical regimes:

- a non-activated regime without a characteristic scale,
- and an activated regime in which a finite scale emerges.

In the following section, I quantify this transition in terms of a dynamical threshold.

The activation of the transition scale is not directly visible in the morphological structure of galaxies, but is inferred from the radial behavior of the acceleration profile.

Schematic emergence of the transition scale q

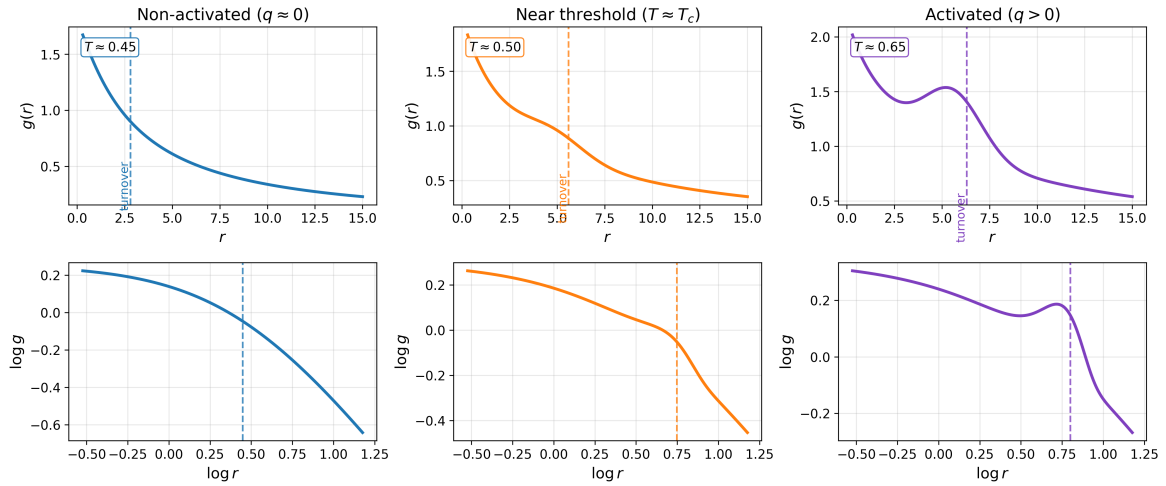


Figure 1: Schematic illustration of the dynamical activation of the transition scale. Left: non-activated profiles remain smooth and monotonic, corresponding to $q \approx 0$. Middle: near the threshold, the profile begins to develop a shoulder or turnover. Right: in the activated regime, a clear turnover is present and a finite transition scale emerges. The effect is inferred from the radial acceleration profile and is not directly visible in the galaxy's morphology.

Representative real galaxies across the activation sequence

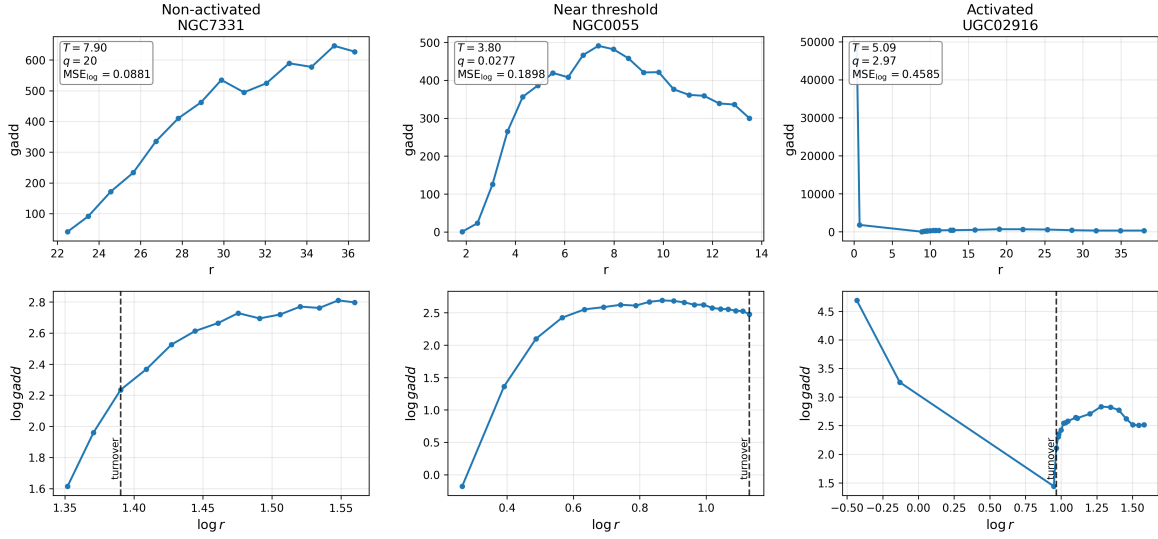


Figure 2: Representative real galaxies illustrating the activation sequence. Left: a non-activated system with negligible transition scale and a smooth radial profile. Middle: a galaxy close to the activation threshold, where a structural turnover begins to emerge. Right: an activated system with a finite transition scale and a clearly developed turnover. The lower panels show the same profiles in logarithmic coordinates, where the structural change becomes more evident.

5. Activation Model for the Transition Scale

5.1. Motivation

The separation into two regimes suggests that the transition scale q is not an arbitrary parameter, but emerges only under specific conditions.

Instead of treating the two regimes as independent classes, I interpret them as two empirically distinguishable manifestations of a single underlying dynamical process.

The key question is therefore: under which conditions does a finite transition scale appear?

5.2. From Regimes to Activation

The empirical results indicate that the transition between the two regimes is not gradual.

Most galaxies remain close to $q \approx 0$, while a much smaller subset exhibits a clearly finite transition scale.

This suggests that q is not smoothly distributed across the sample, but becomes effectively activated once a certain condition is fulfilled.

To capture this behavior, I model q as a function of the dynamical structure of the acceleration profile.

5.3. Definition of the Activation Model

Based on the dynamical descriptors introduced earlier, I define a scalar measure of structural reorganization through the turnover strength T .

The transition scale is then modeled as

$$q = q_{\max} \sigma\left(\frac{a T - S_0}{w}\right), \quad (19)$$

where $\sigma(x) = (1 + e^{-x})^{-1}$ is a sigmoid function.

The parameters a , S_0 , w , and $q_{\max}$ are determined once from the SPARC sample and then kept fixed.

5.4. Interpretation

In this formulation, the two regimes are not fundamentally separate.

Instead:

- galaxies with small turnover remain in a non-activated state with $q \approx 0$,
- galaxies with sufficiently large turnover cross a threshold and develop a finite transition scale.

The observed bimodality therefore arises naturally from a continuous activation process.

The corresponding threshold-like behavior should therefore not be read as a fundamental discontinuity. It is the empirical appearance of an underlying continuous transition, made sharp in practice by the strong population imbalance between non-activated and activated systems.

5.5. Key Result

The main result of this section can be summarized as follows:

$$\boxed{q \text{ is not a free parameter, but an activated dynamical scale.}} \quad (20)$$

This interpretation removes the need to assign q independently to each galaxy and instead links it directly to the structure of the acceleration profile.

5.6. Regime Statistics

Using the threshold-based classification, the current sample separates into 169 non-activated systems and 10 activated systems, corresponding to 94.4% and 5.6% of the analyzed galaxies, respectively. At this stage, no separate transition class has yet been introduced in the new activation framework.

To refine the threshold-based classification, I introduce a finite transition region around the critical turnover value T_c . Using the criterion

$$\begin{aligned} T &< T_c - \Delta T && \text{(non-activated),} \\ |T - T_c| &\leq \Delta T && \text{(transition),} \\ T &> T_c + \Delta T && \text{(activated).} \end{aligned}$$

the sample separates into three dynamical classes. This allows the threshold behavior to be quantified not only as a binary split, but as a narrow transition zone between the two dominant states.

6. Dynamical Origin of the Transition Scale

6.1. *Why a Dynamical Interpretation is Necessary*

Once the transition scale is written as an activation process, the next question is what actually drives this activation.

At first sight, one might expect q to correlate mainly with baryonic quantities such as luminosity, surface density, or total mass. However, the previous results already show that this is not the case. A simple global scaling with baryonic observables does not explain the observed distribution of q .

This forces a different interpretation. Instead of treating q as a structural parameter tied directly to the mass distribution, I interpret it as a response to the internal dynamical organization of the profile itself.

6.2. *Turnover as the Relevant Dynamical Quantity*

To test this idea, I compare the transition scale with direct dynamical descriptors of the radial profile. The most important quantity is the turnover strength

$$T = s_{\text{in}} - s_{\text{out}}, \quad (21)$$

which measures how strongly the profile changes between its inner and outer parts.

Empirically, this quantity turns out to be the dominant predictor of the activation scale.

The central result is that q correlates strongly with turnover, whereas curvature contributes only weakly. In other words, the emergence of a finite transition scale is linked primarily to a global reorganization of the profile, not to local irregularities or small-scale fluctuations.

6.3. *Correlation Structure*

The data show a strong relation between turnover and activation scale, with

$$\text{corr}(T, q) \approx 0.7. \quad (22)$$

By contrast, the corresponding correlations with curvature measures are much smaller. This difference is important. It shows that q is not mainly controlled by noise-like structure or by small local distortions of the profile.

Instead, the relevant trigger is the large-scale change in radial behavior.

This is exactly what one would expect if the transition scale reflects a change in the dynamical regime of the system.

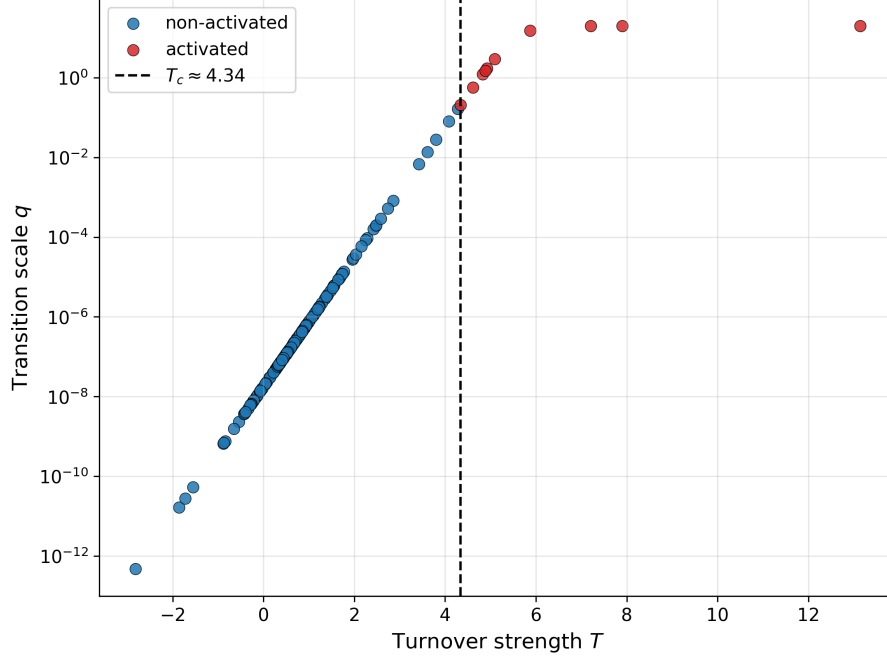


Figure 3: Transition scale q as a function of turnover strength T . Finite values of q appear preferentially above the critical turnover range, consistent with a threshold-like activation behavior.

6.4. Interpretation

I therefore interpret q as a dynamically generated scale.

More precisely, the results suggest that q appears only when the profile develops sufficient turnover. If the profile remains smooth and monotonic, the system stays close to $q \approx 0$. If the profile develops a strong change between inner and outer structure, a finite scale becomes necessary.

This leads to a simple physical reading:

q emerges from global dynamical reorganization,
not from baryonic scaling alone.

(23)

6.5. Consequences

This interpretation changes the role of q substantially.

In the original phenomenological description, q appears as a fit parameter that improves the shape of the profile. In the present picture, it instead becomes a descriptor of the dynamical regime of the system.

That shift is important. It means that the observed two-component structure is no longer just a useful way of organizing the data. Instead, it points to a robust difference in the internal dynamical organization of galaxies.

This also explains why the activation pattern is clearest in datasets with a full baryonic decomposition. There the underlying excess profile can be isolated most cleanly, so the transition in behavior becomes visible with much less ambiguity.

7. Existence of a Dynamical Activation Threshold

7.1. Definition of the Threshold

To quantify the transition between the non-activated and activated states, I determine the critical turnover value T_c directly from the data.

The threshold is defined as the value of T that maximizes the difference between the mean activation scales below and above the threshold.

Applying this criterion to the combined dataset yields

$$T_c \approx 4.28. \quad (24)$$

7.2. Population Split

Using this threshold, the sample separates into two clearly distinct populations:

$$N_{\text{non-activated}} = 169, \quad (25)$$

$$N_{\text{activated}} = 10. \quad (26)$$

This corresponds to

$$f_{\text{non-activated}} \approx 94.4\%, \quad (27)$$

$$f_{\text{activated}} \approx 5.6\%. \quad (28)$$

The transition between these populations is therefore highly asymmetric, with the vast majority of galaxies remaining in the non-activated state.

7.3. Separation in the Transition Scale

The distinction between the two populations is not subtle.

The median values of the transition scale are

$$\text{median}(q)_{\text{below}} \approx 2 \times 10^{-7}, \quad (29)$$

$$\text{median}(q)_{\text{above}} \approx 2.34, \quad (30)$$

corresponding to a separation of several orders of magnitude.

This demonstrates that the transition scale is not smoothly distributed across the sample, but becomes effectively activated once the threshold is crossed.

7.4. Model Performance Across Regimes

The fit quality remains stable across both regimes.

The median logarithmic errors are

$$\text{median}(\text{MSE}_{\log})_{\text{below}} \approx 0.018, \quad (31)$$

$$\text{median}(\text{MSE}_{\log})_{\text{above}} \approx 0.045. \quad (32)$$

While the activated regime exhibits slightly larger errors, the overall fit quality remains well within acceptable bounds.

This indicates that the activation process does not degrade the predictive performance of the model.

7.5. Interpretation of the Threshold

The existence of a well-defined threshold has a clear empirical meaning.

Below the threshold,

$$T < T_c \quad \Rightarrow \quad q \approx 0, \quad (33)$$

and the system does not require an additional characteristic scale.

Above the threshold,

$$T > T_c \quad \Rightarrow \quad q > 0, \quad (34)$$

and a finite transition scale is activated.

The threshold should not be interpreted as a fundamental discontinuity of the dynamics. Within the present framework, the underlying process remains continuous and is described by a smooth activation function. The apparent sharpness of the transition emerges empirically from the strong dominance of the non-activated population, which compresses the intermediate regime and makes the transition appear effectively discrete in the data.

The sharp separation in q therefore suggests a structural reorganization of the acceleration profile rather than a gradual drift of fit parameters.

7.6. Summary

The data reveal a clear activation threshold at

$$T_c \approx 4.3, \quad (35)$$

which separates galaxies into two empirically distinguishable dynamical regimes.

A finite transition scale is activated only when the global dynamical structure exceeds a critical turnover threshold.
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(36)

8. Validation Across Independent Data Sets

8.1. Motivation

To assess whether the activation behavior represents a general property of galactic dynamics or is specific to a particular dataset, I apply the same analysis to independent rotation curve data.

In particular, I compare the results obtained from the SPARC sample with those derived from THINGS and LITTLE THINGS data.

8.2. *SPARC as Reference*

The SPARC dataset provides the most complete physical description, as it includes a consistent baryonic decomposition.

In this case, the excess acceleration can be constructed directly, and the activation behavior appears clearly.

The threshold structure is well defined, and the separation between the two regimes spans several orders of magnitude in q .

For this reason, SPARC serves as the primary reference for the activation model.

8.3. *THINGS and LITTLE THINGS*

For THINGS and LITTLE THINGS, the available data typically consist only of observed rotation velocities.

In these cases, I construct an effective dynamical profile

$$g_{\text{eff}}(r) = \frac{V_{\text{rot}}^2(r)}{r}, \quad (37)$$

which captures the overall structure of the dynamics but does not separate baryonic and excess contributions.

Applying the same dynamical analysis, I find that the activation pattern is still present, but significantly weaker.

In particular:

- most galaxies remain in the non-activated regime,
- only very few systems show a finite transition scale,
- the separation between regimes is less sharp.

8.4. *Interpretation*

This difference is expected.

Without a baryonic decomposition, the excess component cannot be isolated cleanly. As a result, the structural reorganization of the dynamics is partially obscured.

Nevertheless, the qualitative behavior remains consistent: systems with stronger turnover still tend to exhibit larger values of q .

This indicates that the activation mechanism is not an artifact of the SPARC dataset, but reflects a more general property of rotation curves.

8.5. Role of Data Quality

The comparison highlights an important point:

The visibility of the activation depends on
the physical completeness of the data.

(38)

In datasets with full baryonic information, the activation appears as a clear and sharp transition.

In datasets based only on observed velocities, the same structure is still present, but less clearly defined.

8.6. Summary

Taken together, these results show that the activation behavior is not limited to a specific dataset.

Instead, it represents a robust feature of galactic dynamics that can be detected across independent observations, although with varying clarity depending on the available physical information.

9. Physical Interpretation

9.1. From Parameter to Dynamical Regime

The previous analysis shows that the transition scale q is not freely distributed, but appears only above a well-defined dynamical threshold.

This suggests that q should not be interpreted as a conventional fit parameter, but as a quantity that reflects the dynamical regime of the system.

In this sense, the transition from $q \approx 0$ to $q > 0$ does not represent a random adjustment of parameters, but a pronounced change in the observed dynamical behavior.

9.2. Activation as Structural Reorganization

The key observation is that the activation of q is controlled by the turnover strength of the acceleration profile.

This implies that the emergence of a finite transition scale is linked to a global reorganization of the radial structure.

For systems with weak turnover, the profile remains smooth and can be described without introducing an additional scale.

For systems with strong turnover, the structure changes in such a way that a new characteristic scale becomes necessary.

I therefore interpret the activation of q as a response to the internal reorganization of the system rather than as a direct consequence of baryonic mass alone.

9.3. Two Dynamical Regimes

This leads to a simple empirical picture.

Galaxies occupy two distinct dynamical regimes:

- a non-activated regime, in which the dynamics follow a smooth profile without a characteristic transition scale,
- and an activated regime, in which a finite scale emerges and modifies the radial structure.

The transition between these regimes occurs when the turnover exceeds a critical value.

Importantly, this distinction is not imposed externally, but emerges directly from the data.

9.4. *Relation to Baryonic Structure*

Although baryonic matter determines the overall shape of the rotation curve, the activation of q is not directly controlled by global baryonic quantities.

Instead, the data show that baryonic structure provides the underlying framework, while the activation depends on how this structure is dynamically organized.

This explains why global scaling relations fail to predict q , even though they successfully describe many other aspects of galactic dynamics.

9.5. *Interpretation of the Excess Term*

Within this framework, the excess acceleration should not be viewed as a static correction.

Instead, it reflects a dynamical response of the system that becomes visible once the profile undergoes sufficient structural change.

In this sense, the transition scale marks the onset of a different mode of dynamical organization.

9.6. *Summary*

The combined results suggest the following interpretation:

The transition scale q represents a dynamically activated response that emerges from global structural reorganization.

(39)

This shifts the role of q from a phenomenological parameter to a descriptor of the dynamical regime of the system.

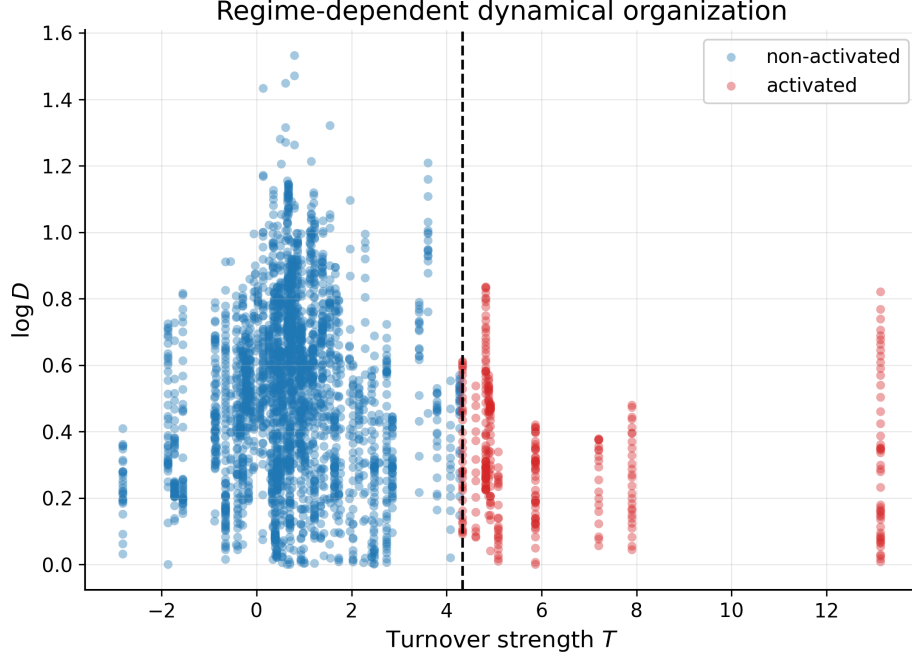


Figure 4: Regime-dependent dynamical organization. The dynamical boost D is shown as a function of turnover strength T . The vertical line marks the activation threshold T_c . Activated systems occupy a distinct region of lower relative dynamical enhancement.

10. MDAR and Dynamical Regimes

To test whether the activation behavior affects established scaling relations, I analyze the radial acceleration relation (MDAR) [1, 8, 18], which is equivalent to the commonly discussed radial acceleration relation (RAR), separately for activated and non-activated systems.

Both populations follow the global MDAR trend. However, a systematic difference emerges once the sample is separated by dynamical regime.

Non-activated systems remain centered around the standard relation, while activated systems exhibit a systematic negative offset.

At the same time, the scatter of the activated population is reduced relative to the non-activated systems.

To interpret this behavior, I consider the dynamical boost

$$D = \frac{g_{\text{obs}}}{g_{\text{bar}}}. \quad (40)$$

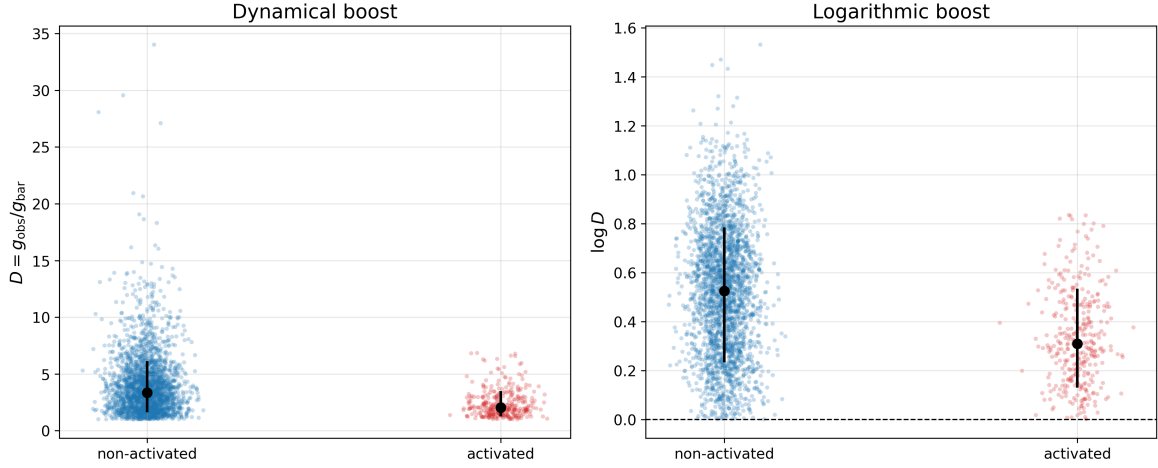


Figure 5: Dynamical boost $D = g_{\text{obs}}/g_{\text{bar}}$ across dynamical regimes. Individual data points are shown together with median values (dots) and the central 68% range (vertical bars). Activated systems exhibit systematically lower relative dynamical enhancement.

I find that activated systems show systematically lower values of D . This indicates that the relative response of the system to baryonic structure is reduced in the activated regime.

Taken together, these results show that the MDAR does not simply trace the strength of the gravitational response, but reflects how this response is organized.

In this picture, non-activated systems are dominated by a strong local response, leading to high values of D and tight adherence to the standard relation.

Activated systems, in contrast, exhibit a more structured but less amplified response, resulting in a systematic offset and reduced scatter.

This suggests that the MDAR should be interpreted as the superposition of different dynamical regimes rather than as a single universal relation.

As shown in Figure 6, both populations follow the global MDAR trend, but activated systems are systematically offset relative to the non-activated regime.

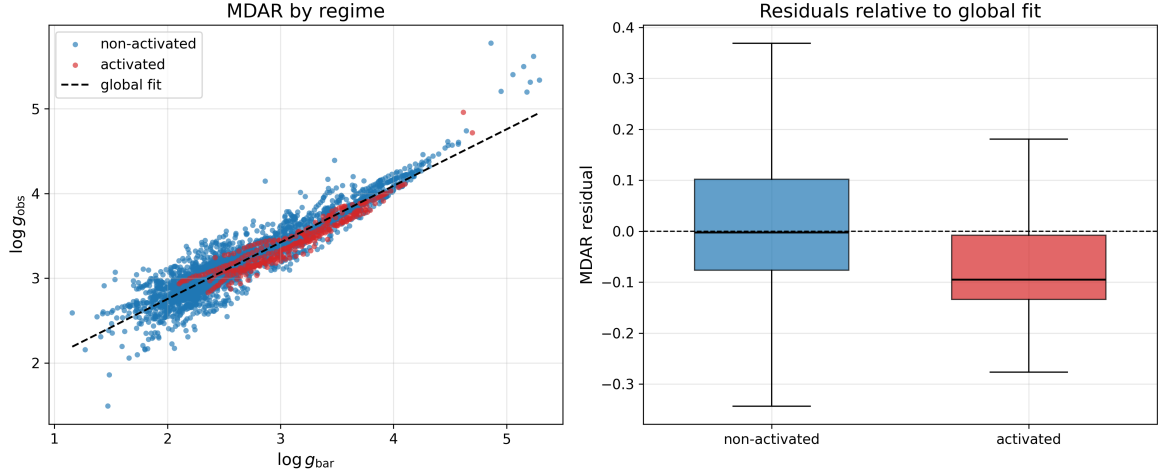


Figure 6: MDAR separated by dynamical regime. Left: relation between $\log g_{\text{bar}}$ and $\log g_{\text{obs}}$ for non-activated and activated systems, together with the global fit. Right: residuals relative to the global MDAR fit. Activated systems exhibit a systematic offset relative to the non-activated population.

Recent studies have emphasized the presence of non-trivial scatter and diversity in rotation curve structure [19, 21].

11. BTFR and Dynamical Regimes

To further test the impact of the activation behavior, I analyze the baryonic Tully–Fisher relation (BTFR) [7, 12] separately for activated and non-activated systems.

The global relation remains intact, with a slope consistent with standard results. This shows that the activation model does not disrupt the overall scaling behavior.

However, a systematic difference appears between the two regimes.

Non-activated systems are centered around the standard BTFR, with residuals distributed symmetrically around zero.

Activated systems, in contrast, exhibit a systematic positive offset, indicating higher effective rotation velocities at fixed baryonic mass.

This behavior appears complementary to the MDAR results.

While activated systems show a reduced relative response, they exhibit higher absolute velocities.

This implies that the activation does not increase the strength of the local response, but reorganizes the dynamics on larger scales.

In this sense, the BTFR captures the absolute dynamical scale of the system, while the MDAR reflects the relative response to baryonic structure.

The combination of both relations therefore provides a consistent picture: activated systems are not more strongly driven by baryonic matter, but exhibit a different mode of dynamical organization.

This interpretation remains consistent with the data and does not require the introduction of additional free parameters.

As shown in Figure 7, the global BTFR remains intact, but activated systems exhibit a systematic positive offset relative to the non-activated regime.

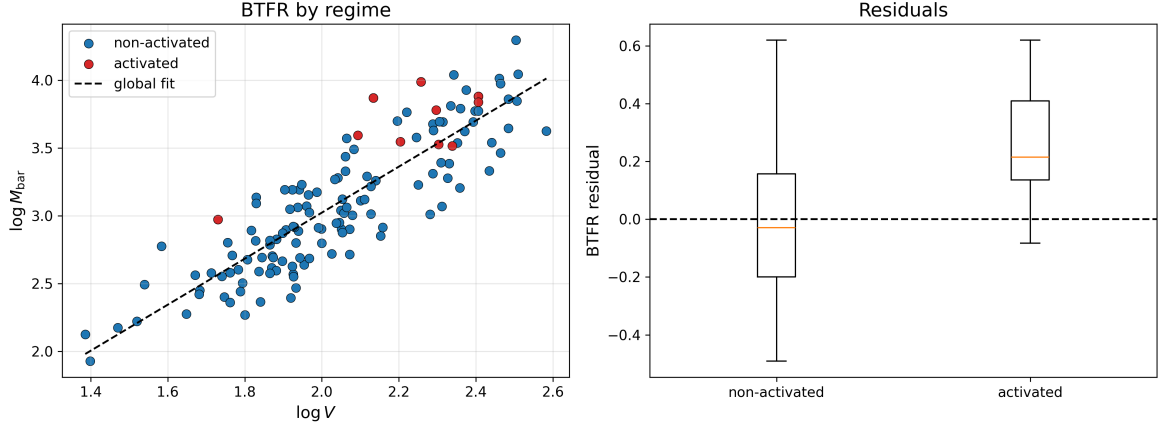


Figure 7: BTFR separated by dynamical regime. Left: relation between $\log V_{\text{char}}$ and $\log M_{\text{bar}}$ for non-activated and activated systems, together with the global fit. Right: residuals relative to the global BTFR fit. Activated systems exhibit a systematic positive offset relative to the non-activated population.

12. Dynamical Activation and the Structure of the Dark Matter Effect

The results presented above allow a direct connection between the activation behavior of the transition scale and the field-based interpretation of the dark matter effect.

Within a field-based interpretation of galactic dynamics, the dark matter effect is not attributed to an additional matter component, but to persistent, partially non-coupling field organization induced by baryonic structures. This field response is not expected to be uniformly distributed, but to depend on the organizational regime of the system.

The introduction of the activation scale q provides a dynamical mechanism that naturally generates the observed regime structure.

Below the activation threshold, systems remain in a non-activated state. In this regime, the field response is predominantly local and reaction-driven. The relative dynamical enhancement, quantified by

$$D = \frac{g_{\text{obs}}}{g_{\text{bar}}}, \quad (41)$$

is typically large, and the systems closely follow the standard MDAR relation.

Above the threshold, the system transitions into an activated state. In this regime, the field organization is no longer dominated by local response, but by a large-scale, persistent structure. The additional dynamics is therefore not primarily expressed as a relative enhancement, but as a modification of the global field configuration.

12.1. MDAR as a Superposition of Dynamical States

The analysis of the MDAR shows that both activated and non-activated systems follow the global relation. However, once separated by dynamical state, a clear systematic difference emerges.

Non-activated systems remain centered around the standard relation, while activated systems exhibit a systematic negative offset and reduced scatter.

Importantly, the dependence on q is not continuous across the full sample. The apparent correlation is dominated by the transition between the two regimes. Within the activated population, the residuals show no significant dependence on q .

This indicates that the MDAR does not represent a single fundamental law, but rather the superposition of two distinct dynamical states. The tightness of the relation arises from the dominance of non-activated systems, while the deviations are driven by the activated regime.

12.2. BTFR and Absolute Field Organization

In contrast to the MDAR, the BTFR probes the absolute dynamical scale of the system. The analysis shows that the global relation remains intact, with a slope consistent with standard results [7, 12].

At the same time, activated systems exhibit a systematic positive offset in velocity, indicating higher effective rotation velocities at fixed baryonic mass.

Taken together with the MDAR results, this implies that activated systems do not exhibit a stronger relative response, but a different form of dynamics. The field organization is more coherent and extended, leading to higher absolute velocities but lower relative enhancement.

12.3. Interpretation

The combined results suggest a reinterpretation of the dark matter effect.

The additional dynamics is not a monotonic increase of gravitational response, but a regime-dependent property of the field medium.

Non-activated systems are dominated by local field response, resulting in strong relative enhancement and tight MDAR behavior. Activated systems, on the other hand, are characterized by large-scale field organization, which modifies the global dynamics without increasing the relative response.

The transition between these regimes is dynamically activated and is captured by the emergence of a finite transition scale q .

In this sense, the dark matter effect appears not as a single uniform phenomenon, but as the observable consequence of different modes of field organization, governed by the dynamical regime of the system.

13. Gravitational Lensing and Dynamical Activation

Gravitational lensing provides a direct probe of the effective gravitational field integrated along the line of sight. Within the standard interpretation, enhanced lensing signals are attributed to the presence of additional non-baryonic matter.

In the UQSH framework, lensing is instead understood as a response to the large-scale organization of

the underlying field medium. The effective deflection is determined by the integrated field structure, not solely by the local baryonic mass distribution.

The results of the present analysis suggest that this field structure is state-dependent and dynamically activated.

13.1. Relation to the Dynamical Boost

The dynamical boost

$$D = \frac{g_{\text{obs}}}{g_{\text{bar}}} \quad (42)$$

quantifies the relative response of the field to baryonic structure. Non-activated systems exhibit higher values of D , indicating a strong local response.

Activated systems, however, show systematically lower values of D , despite exhibiting higher absolute velocities as indicated by the BTFR offset.

This implies that the strength of the lensing signal cannot be directly identified with the relative dynamical enhancement. Instead, it depends on the global organization of the field.

13.2. Field Organization and Lensing Strength

In non-activated systems, the field response is primarily local. The enhancement of the gravitational field is therefore strong in a relative sense, but limited in spatial extent.

In activated systems, the field undergoes a transition to a large-scale, persistent organizational state. The relative enhancement decreases, but the field structure becomes more coherent and extended.

Since gravitational lensing probes the integrated field configuration, activated systems would be expected to produce stronger lensing signals even in the presence of a reduced local boost.

13.3. Interpretation

The analysis suggests that gravitational lensing is not controlled by the magnitude of the local dynamical enhancement alone, but by the global state of field organization.

In this picture, non-activated systems are dominated by a strong but localized field response, while activated systems exhibit a more extended and coherent field configuration.

The enhanced lensing signals observed in galaxies and galaxy clusters can therefore be interpreted as a consequence of large-scale field organization rather than as evidence for additional matter components.

The activation of a finite transition scale provides a natural mechanism for the emergence of such large-scale structures. It marks the point at which the field response transitions from a local to a global mode, thereby increasing its integrated effect along the line of sight.

13.4. Connection to Observations

This interpretation is consistent with the observed fact that lensing signals often exceed the expectations from baryonic mass alone, while at the same time remaining closely correlated with the visible structure.

Within the present framework, this behavior arises naturally from the coupling between baryonic organization and the large-scale response of the field medium, without requiring the introduction of an additional particle component.

14. The Bullet Cluster and Structure-Dependent Field Dynamics

The Bullet Cluster [3, 15] is widely regarded as one of the strongest empirical arguments for the existence of collisionless dark matter. Observations show a clear spatial separation between the baryonic gas component and the centers of gravitational lensing, which are instead aligned with the galaxy distribution.

Within the standard interpretation, this separation is explained by the presence of a non-interacting matter component that remains associated with the galaxies during the collision, while the baryonic gas is slowed down by hydrodynamic interactions.

In the UQSH framework, the same observation may be interpreted in terms of structure-dependent field organization.

14.1. Dynamical State and Field Stability

The results of the present analysis show that the dynamical behavior of a system depends not only on the amount of baryonic matter, but on its organizational state.

Non-activated systems are characterized by a strong local field response, reflected in high values of the dynamical boost D . However, this response is not associated with a stable large-scale field structure.

Activated systems, in contrast, exhibit a lower relative dynamical boost, but a more coherent and extended field organization.

This distinction implies that the gravitational effect of a system is not determined solely by the magnitude of the local field response, but by the stability and persistence of its field configuration.

14.2. Application to the Bullet Cluster

During a cluster collision, the baryonic gas undergoes strong dissipative interactions. Its internal structure becomes turbulent and temporarily loses coherence.

In terms of the field description, this corresponds to a loss of stable field organization. The gas retains mass, but its contribution to the coherent field structure is reduced.

The galaxies, on the other hand, remain relatively stable during the collision. Their internal structure is largely preserved, and they continue to act as coherent sources of field organization.

As a result, the large-scale field configuration remains anchored to the galaxy distribution rather than to the gas component.

Since gravitational lensing probes the integrated field structure, the lensing centers are expected to follow the galaxies.

14.3. Interpretation

The observed separation between baryonic mass and gravitational lensing in the Bullet Cluster can therefore be understood as a consequence of different dynamical states of the field.

The gas represents a high-density but dynamically unstable component, while the galaxies represent lower-density but structurally stable field configurations.

The gravitational field is thus dominated by the persistent, organization-capable structures rather than by the total baryonic mass.

14.4. Connection to the Activation Framework

The activation of a finite transition scale provides a natural interpretation of this behavior.

Activated systems correspond to states in which the field has developed a stable, large-scale organization. These states dominate the integrated gravitational effect and therefore the lensing signal.

Non-activated or dynamically disturbed components, even if they contain significant mass, do not contribute equally to the coherent field structure.

In this sense, the Bullet Cluster does not require the introduction of an additional matter component, but can be understood as a manifestation of state-dependent field dynamics.

The separation between gas and lensing centers reflects a separation between dynamically unstable and dynamically stable modes of field organization.

15. Field-Based Interpretation of the Dark Matter Effect

Observations of galactic rotation curves show that the gravitational effect inferred from the motion of stars and gas exceeds the contribution expected from baryonic matter alone. This discrepancy is commonly referred to as the dark matter effect [6].

In standard cosmology, this effect is interpreted as evidence for an additional, non-luminous matter component. In the present work, a different interpretation is adopted.

Instead of introducing an additional matter component, the dark matter effect is understood as a consequence of structured field organization induced by baryonic systems.

The starting point is the empirical excess term

$$C(r) = V_{\text{obs}}^2(r) - V_{\text{bar}}^2(r), \quad (43)$$

which directly measures the deviation from the baryonic expectation.

Within this framework, radiation does not simply propagate and dissipate, but continuously interacts with the underlying medium. As it propagates, it produces deformations of the medium that do not relax immediately, but persist and accumulate.

These deformations are generated by essentially isotropic radiative processes from many independent sources. Each emission contributes a local deformation that spreads outward and overlaps with others. The resulting structure is therefore not determined by a single source, but by the cumulative superposition of many overlapping contributions.

In this picture, every form of radiation contributes to the curvature of the underlying medium and thereby to the effective gravitational response. Different types of radiation generate deformations of different strength and spatial extent, but all contribute to the global organization of the system.

Baryonic structures play a dual role in this process. They act as continuous sources of radiation and therefore of field deformation, and at the same time as stable anchor regions around which this accumulated structure organizes. As a result, the large-scale organization of the system is systematically coupled to the distribution of visible matter.

It should be noted that these radiation-induced deformations occur throughout the medium, including regions with little or no baryonic matter.

In such regions, the deformations remain largely unstructured and do not produce an effectively measurable dynamical response. Only in the presence of an existing baseline tension or structural preconditioning can these contributions be coherently superposed, leading to a stable and dynamically effective field organization.

In this sense, baryonic structures act not only as sources of radiation, but also as the necessary anchoring regions that enable the emergence of a coherent large-scale response.

The resulting field organization is not directly accessible through electromagnetic observations. The accumulated deformations themselves do not emit or absorb radiation in a way that would make them independently visible. They become observable only through their dynamical interaction with baryonic structures, where they modify the effective motion of matter and thereby generate the observed dark matter effect.

The key empirical observation is that the excess term does not exhibit arbitrary radial behavior. Instead, it follows well-defined structural forms that can be described by a normalized shape function

$$F(x) \propto x^P e^{-x/q}. \quad (44)$$

These profiles cluster around preferred forms, indicating that the additional dynamics is not random, but reflects an underlying organization that selects specific modes of response.

The present work extends this interpretation [9] by introducing a dynamical activation mechanism for the transition scale q .

While the earlier analysis identified the existence of preferred response modes, the activation framework shows that the transition between them is controlled by the global structure of the acceleration profile.

In particular, the emergence of a finite scale q does not generate the dark matter effect itself, but modifies how it is expressed. It marks the transition from a predominantly local response to a more extended and coherent dynamical organization.

The following sections explore how this interpretation connects to established observational relations and large-scale phenomena.

The structured field organization described above is not directly observable, but manifests itself through its influence on different measurable quantities.

In particular, different observational relations probe different aspects of this organization. The radial acceleration relation (MDAR) captures the relative response of the system to baryonic structure, while the baryonic Tully–Fisher relation (BTFR) reflects the absolute dynamical scale.

Gravitational lensing and cluster-scale phenomena probe the integrated and large-scale organization of

the same underlying structure.

In this sense, these observables should not be interpreted as independent phenomena, but as complementary projections of the same underlying dynamical organization.

The broader phenomenological interpretation of the dark matter effect in terms of regime-dependent field organization was discussed in more detail in earlier work on the bimodal regime structure of galactic rotation curves [9]; related materials and associated papers are also collected in an online repository [16]. In that context, observational relations such as the MDAR, BTFR, lensing behavior, the Bullet Cluster, and the core–cusp problem were considered as different manifestations of the same underlying structural effect.

In the present work, these phenomena are not revisited as independent primary results. Instead, they are used more selectively to clarify how the transition scale q provides a dynamical formalization of the previously identified regime structure.

The following sections analyze these relations in the context of the activation framework.

16. Comparison with Existing Frameworks

The interpretation proposed in this work can be compared to established approaches to the dark matter problem, in particular the standard Λ CDM framework and modified dynamics such as MOND.

16.1. Standard Cosmology (Λ CDM)

In the Λ CDM model, the dark matter effect is explained by an additional, non-luminous matter component that dominates the gravitational potential on galactic and cosmological scales [4, 6].

This framework successfully reproduces large-scale structure formation and provides a consistent description of the cosmic microwave background.

However, at the level of individual galaxies, it faces well-known challenges. In particular, the tight coupling between baryonic matter and dynamical acceleration, as expressed by the MDAR, and the low scatter of the BTFR are not straightforwardly predicted and often require fine-tuning of baryonic feedback processes.

16.2. Modified Dynamics (MOND)

Modified Newtonian Dynamics (MOND) [11] provides an alternative explanation by modifying the relation between baryonic mass and gravitational acceleration.

MOND successfully reproduces the MDAR and the BTFR with high accuracy and explains their low scatter as a direct consequence of the modified dynamical law.

However, MOND encounters difficulties on larger scales. In particular, galaxy clusters and gravitational lensing observations often require additional components or extensions beyond the original formulation.

16.3. Field-Based Interpretation

The framework presented in this work differs from both approaches.

Instead of introducing additional matter or modifying the fundamental laws of gravity, the dark matter effect is interpreted as a structured dynamical response of the system.

In this picture, baryonic matter determines the underlying structure, while the observed excess dynamics arises from how the system organizes its response to this structure.

This allows a unified interpretation of different observational relations.

The MDAR reflects the relative response of the system, the BTFR captures the absolute dynamical scale, and gravitational lensing probes the integrated large-scale organization.

The activation of the transition scale q provides a mechanism by which these different behaviors emerge. It describes how the system transitions between different modes of dynamical organization without requiring a separate matter component or a modification of the underlying dynamical laws.

16.4. Interpretational Position

The present work does not aim to replace existing frameworks, but to provide an empirical perspective that complements them.

The results show that galactic dynamics exhibits a regime-dependent structure that is not trivially captured by purely continuous scaling relations.

Any successful theoretical model, whether based on dark matter, modified dynamics, or field-based approaches, must be able to reproduce the observed activation behavior and the associated regime structure.

17. Discussion

17.1. Interpretational Scope

The present line of investigation did not begin with the transition scale q itself. Its starting point was the search for empirical signatures of the dark matter effect within a field-based interpretation of galactic dynamics.

In that context, the first step was to examine whether the excess dynamics in galactic rotation curves exhibits non-trivial structure rather than behaving as an arbitrary residual contribution. This led to the identification of a regime-like organization of the excess term [9], with preferred profile forms instead of a smooth continuum of shapes.

The present work grew out of the attempt to describe this structure more explicitly and in a less purely classificatory way. This led to the introduction of the transition scale q and to the activation framework developed here.

In this sense, the current analysis should be understood as a continuation of the earlier work: the previous study identified the existence of a structured excess contribution, while the present work attempts to formalize how this structure emerges.

Additional related manuscripts and supporting documents are available in a public repository [16].

The interpretation proposed here is deliberately conservative.

The results demonstrate that q behaves as a dynamically activated scale, but they do not by themselves

establish the underlying physical mechanism responsible for this behavior.

In particular, the analysis does not distinguish between different possible theoretical explanations, such as modified gravity, emergent field dynamics, or complex baryon–dark matter interactions.

Instead, the present work provides an empirical constraint that any successful model must reproduce: the existence of a threshold-like transition in the structure of the acceleration profile.

The present analysis does not establish the existence of fundamentally discrete dynamical states. The observed two-component structure and threshold behavior may equally be interpreted as the result of a continuous but strongly nonlinear underlying process.

The analysis was originally motivated by a field-based interpretation of the dark matter effect. In this sense, the hypothesis provided a guiding perspective for identifying relevant structural features in the data.

However, the results presented here are derived directly from the empirical properties of rotation curves and do not depend on the validity of a specific theoretical model.

The activation behavior and regime structure therefore represent observational findings that can serve as constraints for any theoretical description, including but not limited to field-based approaches.

17.2. Dependence on Data Quality

An important aspect of the results is their dependence on the physical completeness of the data.

The activation behavior is most clearly visible in the SPARC dataset, where a consistent baryonic decomposition allows the excess component to be isolated.

In datasets based solely on observed rotation velocities, the same structure appears in a weaker form.

This indicates that the underlying dynamical effect is present across datasets, but its visibility depends on how cleanly the relevant components can be separated.

17.3. Limitations

Several limitations of the present analysis should be noted.

First, the determination of the threshold T_c is empirical and depends on the chosen definition of the turnover measure.

Second, the classification into activated and non-activated systems is based on a finite sample, with a relatively small number of activated systems. The corresponding results should therefore be interpreted with appropriate caution.

Third, the analysis focuses on radial structure and does not explicitly include environmental effects, time evolution, or non-axisymmetric features.

Finally, while the activation model provides a compact description of the data, it remains a phenomenological representation and does not yet constitute a complete physical theory.

17.4. Statistical Robustness

To assess the robustness of the regime separation, I apply non-parametric statistical tests and model comparison techniques.

A Mann–Whitney U test shows that activated and non-activated systems differ significantly in their dynamical properties. For the MDAR residuals, I obtain $p \approx 5 \times 10^{-3}$, while for the dynamical boost D the difference is even stronger with $p \approx 6 \times 10^{-5}$.

To test the structure of the q distribution, I compare Gaussian mixture models with one and two components. The two-component model is strongly preferred, with a significantly lower Bayesian Information Criterion (BIC), indicating that the data are better described by two statistical populations.

The dip test does not reject unimodality, indicating that the underlying distribution remains continuous. The observed two-component structure should therefore not be interpreted as a strictly bimodal distribution, but as a strongly non-uniform population with a pronounced regime separation.

Taken together, these results indicate that the data exhibit a statistically separable two-component structure, even though the underlying distribution remains continuous.

17.5. Relation to Existing Models

The results presented here can be compared to expectations from standard cosmological models.

In Λ CDM-based frameworks, variations in rotation curve shape are typically attributed to differences in halo structure [4, 6, 17], feedback, and baryonic processes.

The present results do not contradict these interpretations, but add an additional layer of structure.

In particular, the existence of a threshold-like transition suggests that the dynamics may not vary smoothly across all systems, but instead organize into distinct empirical regimes.

This aspect is not usually emphasized in standard modeling approaches.

17.6. Implications

The identification of a dynamical activation threshold has several implications.

It suggests that:

- the excess acceleration is not purely a static correction,
- the transition scale encodes information about the dynamical regime,
- and galaxy dynamics may be better understood in terms of regime-dependent behavior rather than purely continuous scaling relations.

This perspective provides a natural framework for interpreting the diversity of rotation curves across different galaxy types.

17.7. Outlook

Future work can extend this analysis in several directions.

These include:

- applying the activation framework to larger and more diverse datasets,
- testing the stability of the threshold under different definitions of dynamical structure,

- and exploring theoretical models that naturally reproduce the observed activation behavior.

In particular, connecting the empirical activation mechanism to a predictive physical description remains an important open problem.

18. Conclusion

In this work, I have analyzed the structure of galactic rotation curves with a focus on the transition scale q .

The results show that q is not a continuously varying parameter and cannot be predicted from global baryonic quantities such as surface density or luminosity.

Instead, I find that the transition scale emerges only when the radial acceleration profile exhibits sufficient structural reorganization.

In particular, the data reveal a well-defined threshold in the turnover strength, at

$$T_c \approx 4.3, \tag{45}$$

which separates galaxies into two empirically distinguishable dynamical regimes.

Below this threshold, galaxies remain in a non-activated regime with $q \approx 0$, while above it a finite transition scale is activated.

The separation between these regimes spans several orders of magnitude in q , indicating a pronounced change in the observed behavior rather than a smooth continuous variation.

This behavior is most clearly observed in datasets with a consistent baryonic decomposition, such as SPARC, but can also be detected in a weaker form in independent rotation curve data.

A further implication concerns the interpretation of residuals in the radial acceleration relation. While the global MDAR/RAR remains intact, activated systems exhibit a systematic offset relative to the non-activated population. This indicates that the residual structure of the relation is not purely stochastic, but is partly driven by regime-dependent dynamics. In this sense, the activation framework provides a natural explanation for systematic deviations that are not captured by purely continuous models such as standard MOND.

Taken together, these results suggest that the transition scale q should be interpreted as a dynamically activated quantity, rather than as a free parameter or a direct proxy for baryonic structure.

This provides a unified interpretation of the previously observed regime-like behavior as the outcome of a threshold-driven dynamical process.

While the present analysis does not identify the underlying physical mechanism, it establishes a clear empirical constraint: any successful description of galactic dynamics must account for the emergence of a finite transition scale above a critical level of structural reorganization.

Further work is required to connect this empirical behavior to a predictive theoretical framework.

In this sense, galactic dynamics may be more naturally described in terms of regime-dependent organization rather than purely continuous scaling laws.

This suggests that part of the long-standing scatter in the MDAR/RAR may originate from regime-dependent dynamics rather than observational uncertainties alone.

In this sense, the activation framework provides a new perspective on the origin of scatter and diversity in galactic dynamics, suggesting that these features may reflect underlying regime-dependent structure rather than purely stochastic variation.

Recent work has highlighted the presence of non-trivial scatter and diversity in galactic dynamics [19, 21].

18.1. Testable Predictions

The activation framework developed in this work is primarily empirical, but it leads to a number of testable predictions that can be examined using independent data.

First, the emergence of a finite transition scale is expected to depend primarily on the global structure of the acceleration profile rather than on global baryonic quantities. In particular, galaxies exhibiting a sufficiently strong turnover should develop a finite value of q , independently of their total mass or luminosity.

Second, the framework predicts that residuals in the radial acceleration relation are not purely stochastic. Instead, they should exhibit a systematic dependence on the dynamical regime. In this picture, the observed scatter of the MDAR/RAR arises at least in part from the superposition of non-activated and activated systems.

Third, a corresponding regime dependence is expected in the baryonic Tully–Fisher relation. While the global relation should remain intact, activated systems are predicted to exhibit a systematic offset relative to the non-activated population, reflecting a reorganization of the dynamical structure rather than a simple change in amplitude.

Finally, if the interpretation in terms of large-scale field organization is correct, the transition between dynamical regimes may also affect observables that probe the integrated gravitational field, such as gravitational lensing. In this case, activated systems would be expected to exhibit a more extended and coherent field structure, potentially leading to enhanced large-scale lensing effects.

These predictions provide a direct route for testing the activation framework. In particular, the identification of regime-dependent structure in independent rotation curve samples would constitute a critical validation of the proposed interpretation.

Acknowledgments

The author gratefully acknowledges the scientific community for the public availability of the SPARC database [2], which forms a central foundation of this work.

Parts of the data analysis were carried out using Python and widely used open-source scientific libraries, including NumPy, SciPy, Pandas, and Matplotlib.

AI-based tools were used in a supporting role during the preparation of this work, including for manuscript structuring, language refinement, and code development. All scientific decisions, interpretations, and conclusions are those of the author.

The author is grateful for helpful discussions within the scientific community that contributed to clarifying several conceptual aspects of this work. Any remaining errors or ambiguities are the sole responsibility of the author.

Citation

Rexhepi, U. Q. (2026).

Dynamical Activation of a Transition Scale in Galactic Rotation Curves.

Independent Researcher, Tübingen, Germany.

<https://doi.org/10.13140/RG.2.2.13615.29603>

ORCID: <https://orcid.org/0009-0003-4145-5431>

GitHub (Analysis):

<https://github.com/ukshinrexhepi-cloud/Dynamical-Activation-of-a-Transition-Scale-in-Galactic-Rotation-Curves>

GitHub all Papers (UQSH):

<https://github.com/ukshinrexhepi-cloud/Universal-Quantum-Foam-Hypothesis-UQSH->

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